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Q8. Suppose two numbers are stored in register R0 and R1. Write an ALP to verify if their sum is greater than FFH

Soln:

```
            MOV   A, R0        ; move first number to A
            ADD   A, R1        ; add second number to it
            JC    MESSAGE      ; if sum > FFH, CY = 1
            SJMP  NEXT         ; if CY = 0, exit
MESSAGE:    MOV   A, #'Y'      ; if CY = 1, move 'Y' into A
            MOV   P1, A        ; send the message 'Y' through P1
NEXT    :   NOP                ; do nothing
            END                ; end of file
```

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Q9. Write an ALP to put 55H in to the upper RAM locations to 90H to 99h

Soln:

```
        ORG     0
        MOV     A,#55H
        MOV     R2,#10
        MOV     R0,#90H ; access the upper 128 bytes
                          of on-chip RAM
BACK:   MOV     @R0,A   ; use indirect addressing mode
        INC     R0
        DJNZ    R2,BACK ; repeat for all locations
        SJMP    $       ; stay here
        END.
```

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Q10 Assume that on-chip ROM has a message. Write a program to copy it from code space into the upper memory space starting at address 80H. Also, as you place a byte in upper RAM, give a copy to P1

Soln:

```
        ORG     0
        MOV     DPTR,#MYDATA
        MOV     R1,#80H         ; access to the upper memory
B1:     CLR     A
        MOVC    A,@A+DPTR       ; copy from code ROM
        MOV     @R1,A           ; store in upper memory
        MOV     P1,A            ; give a copy to P1
        JZ      EXIT            ; exit if last byte
        INC     DPTR            ; increment DPTR
        INC     R1              ; increment R1
        SJMP    B1              ; repeat until last byte
EXIT:   SJMP    $               ; stay here when finished
;-----------
        ORG     300H
MYDATA  DB      "The Promise of World Peace",0
        END.
```

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Q11. Write an ALP to takes 10 bytes of data from data RAM locations 45H to 54H, add 02 to each of them, and save the result in a data RAM locations 79h down to 70h.

Soln:

```
        MOV   R2,#10    ;R2 is used as the counter
        MOV   R0,#45H   ;R0 points to first source location
        MOV   R1,#79H   ;R1 points to first destination location
BACK:   MOV   A,@R0     ;move source byte to A
        ADD   A,#02     ;add 2 to it
        MOV   @R1,A     ;move it to destination
        INC   R0        ;increment the source pointer
        DEC   R1        ;decrement the destination pointer
                        ;this is because the address of destination is in
                        ;decreasing order
        DJNZ  R2,BACK   ;repeat until counter value is zero
```

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Q12. Write an ALP to load the accumulator with the value 55H and complement the accumulator 700 times

Soln:

```
        MOV  A, #55H      ; A = 55H
        MOV  R3, #10      ; R3 = 10, outer loop count
NEXT:   MOV  R2, #70      ; R2 = 70, inner loop count
AGAIN:  CPL  A            ; complement A register
        DJNZ R2, AGAIN    ; repeat it 70 times
        DJNZ R3, NEXT
```

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Q14. Assume that ROM space starting at 250H contains "INDIA", write an ALP to transfer the bytes into RAM locations starting at 40H.

Soln:

;@ This method uses a counter

```
        ORG     0000
        MOV     DPTR,#MYDATA ; load ROM pointer
        MOV     R0, #40H     ; load RAM pointer
        MOV     R2, #7       ; load counter
BACK    CLR     A            ; A=0
        MOVC    A,@A+DPTR    ; move data from code space
        MOV     @R0,A        ; save it in RAM
        INC     DPTR         ; increment ROM pointer
        INC     R0           ; increment RAM pointer
        DJNZ    R2,BACK      ; loop until counter =0
HERE:   SJMP    HERE

;--------on-chip code space used for storing data
        ORG     250H
MYDATA  DB      "INDIA"
        END.
```

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;(b) This method uses null char for end of string

```
        ORG     0000
        MOV     DPTR,#MYDATA ; load ROM pointer
        MOV     R0,#40H      ; load RAM pointer
BACK :  CLR     A            ; A=0
        MOVC    A,@A+DPTR    ; move data from code space
        JZ      HERE         ; exit if null character
        MOV     @R0,A        ; save it in RAM
        INC     DPTR         ; increment ROM pointer
        INC     R0           ; increment RAM pointer
        SJMP    BACK         ; loop
HERE :  SJMP    HERE

;------- On-chip code space used for storing data
        ORG     250H
MYDATA : DB     "INDIA",0    ; notice null char for
                             ; end of string.
        END
```

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Q15 Write an ALP to get the X (from 0 to 9) values from P1 and send X square values to P2 continuously.

Soln:

```
        ORG     0
        MOV     DPTR, #300H     ;LOAD TABLE ADDRESS
        MOV     A#0FFH          ;A= FF
        MOV     P1, A           ;CONFIGURE P1 INPUT PORT
BACK:   MOV     A, P1           ; GET X
        MOV     A, @A+DPTR      ;GEX SQUARE FROM TABLE
        MOV     P2, A           ;ISSUE IT TO P2
        SJMP    BACK            ; KEEP DOING IT

        ORG     300H
XSQR_TABLE:
        DB      0,1, 4, 9, 16, 25, 36, 49, 64, 81
        END
```

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Q15. Write an ALP to add two 16-bit numbers are FC45h and 02ECh

Soln:

```
CLR   C          ; make CY=0
MOV   A,#45H     ; load the low byte into A
ADD   A,#0ECH    ; add the low byte, now A=31, CY=1
MOV   R0,A       ; save the low byte of sum into A
MOV   A,#02H     ; load the high byte into A
ADDC  A,#0FCH    ; add the high bytes with
                   carry
                 ; 02+FCH+1 = FFH
MOV   R1,A       ; save the high byte of result
                   in R1
```

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